

EDA Case Study

By Samuel Luo & Vidhi Thakore



Income levels

Lower average income

Results in higher default rate



Reasons

Lower availability of funds for loan repayment
Probability of higher volatility in income
eg: daily wage earners

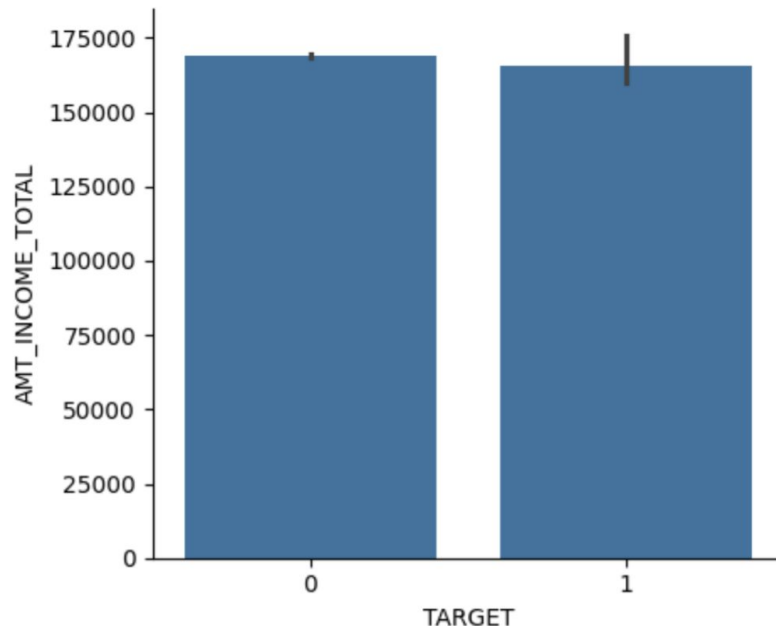


Solution

Providing loans with a more flexible repayment plan & provision of micro loans to cater to lower income individuals



<seaborn.axisgrid.FacetGrid at 0x792f9f6168a0>



Annuity levels

Lower annuity

Results in higher default rate



Reasons

Reduced disposable income & financial cushion

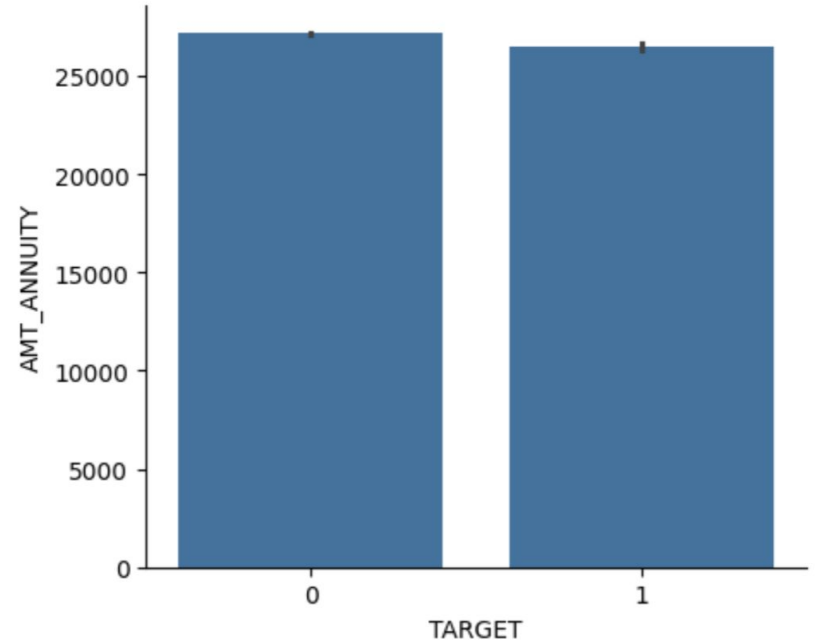


Solution

Limiting the number of loans granted to individuals with lower annuity levels



<seaborn.axisgrid.FacetGrid at 0x792f9e8f7aa0>



Final Credit Amount

Lower Final Credit

Results in higher default rate



Reasons

Indicates bank's ability to identify potential defaulters

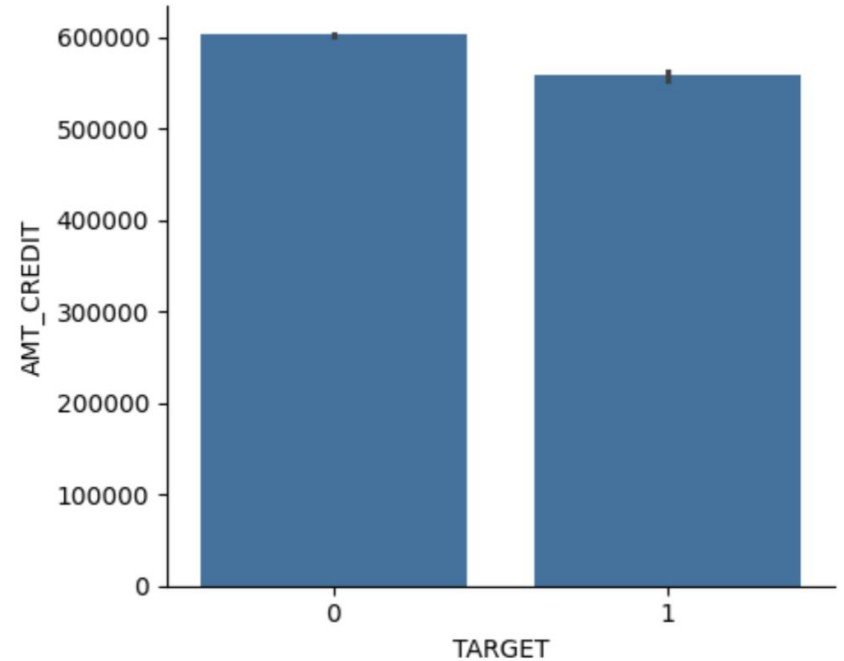


Solution

Minimising number of customers that are given loans after having a lower final credit amount



<seaborn.axisgrid.FacetGrid at 0x792fb42e78f0>



Age

Late 30s

Results in higher default rate



Reasons

Higher responsibilities and expenditures coupled with no/limited increase in income

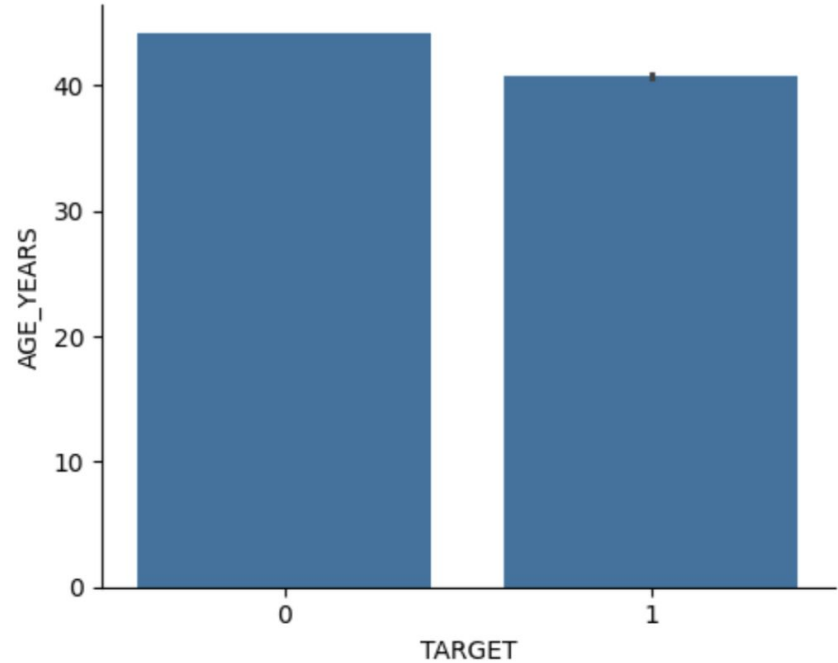


Solution

Providing loans with a more flexible repayment plan



<seaborn.axisgrid.FacetGrid at 0x792fb430bfb0>



Education levels

Lower education level

Results in higher default rate



Reasons

Poor financial planning —lack of access to resources and knowledge
Blue collared/comparatively lower paying jobs—lower disposable income

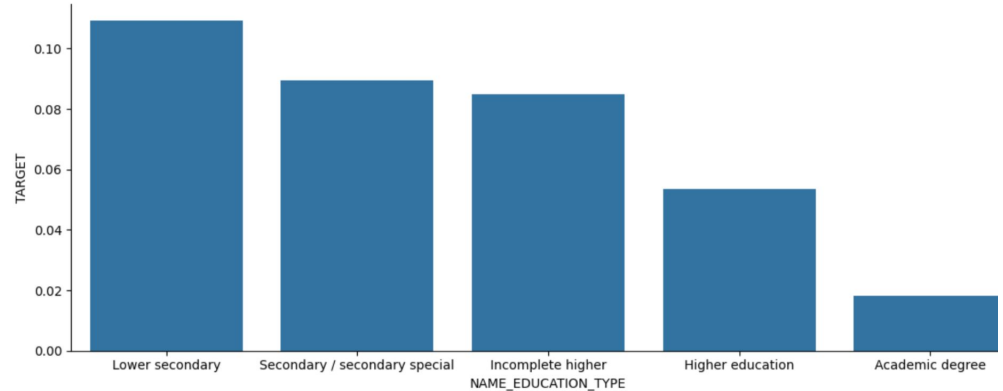


Solution

Providing smaller loans to those with education only until lower secondary



<seaborn.axisgrid.FacetGrid at 0x78bad527f9e0>



Family Status

Civil marriage

Have the highest default rate followed by non-married



Reasons

Lower availability of funds for repayment (single) - no one to rely on in crisis
A partner's financial mishaps affect the other-civil marriage

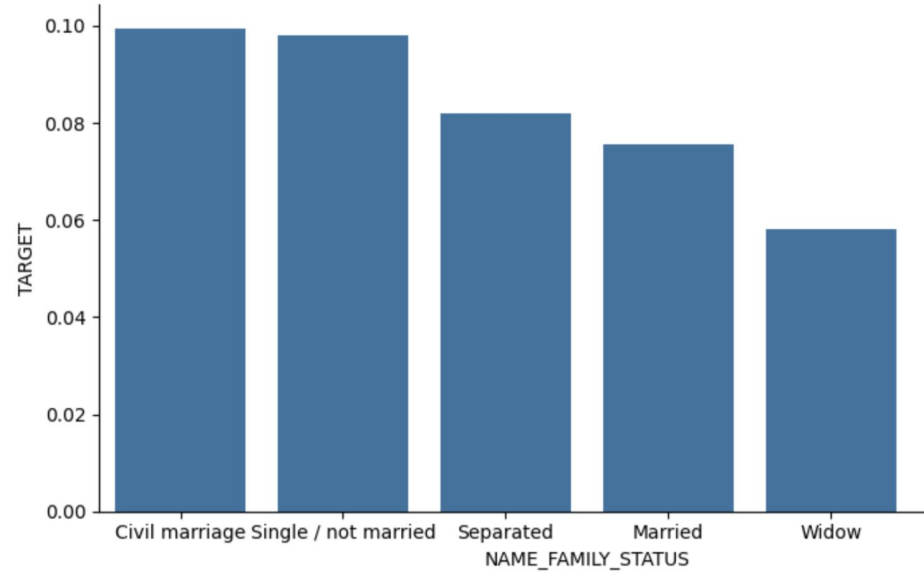


Solution

Gathering data about civil marriage (highest default rate)/ married prior
Ensuring that single individuals don't have reckless spending habits



<seaborn.axisgrid.FacetGrid at 0x792f9e9c21b0>



Income type

Maternity leave

Results in highest default rate



Reasons

Sharp increase in medical expenses,
spending on supplies for the child etc,
alongside a paycut
Shows effectiveness of pension funds

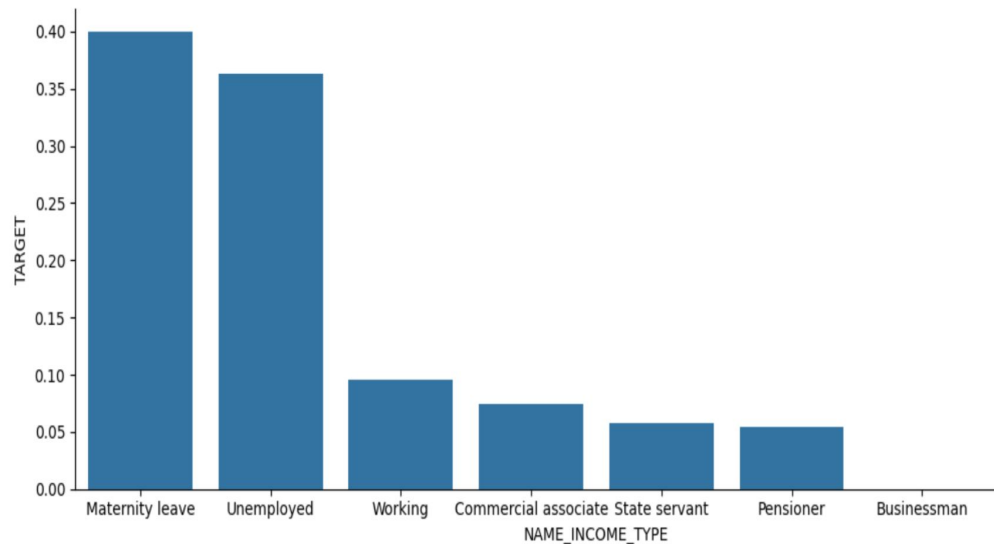


Solution

Prioritising pensioners, state servants,
commercial associates and working
individuals would minimise losses for
the bank



<seaborn.axisgrid.FacetGrid at 0x78bad2ef7590>



Occupation

Low skill laborers

Have the highest default rate
Accountants have the lower default rate



Reasons

Blue collar workers have insufficient and volatile income
Accountants & white collar workers have the skills and income

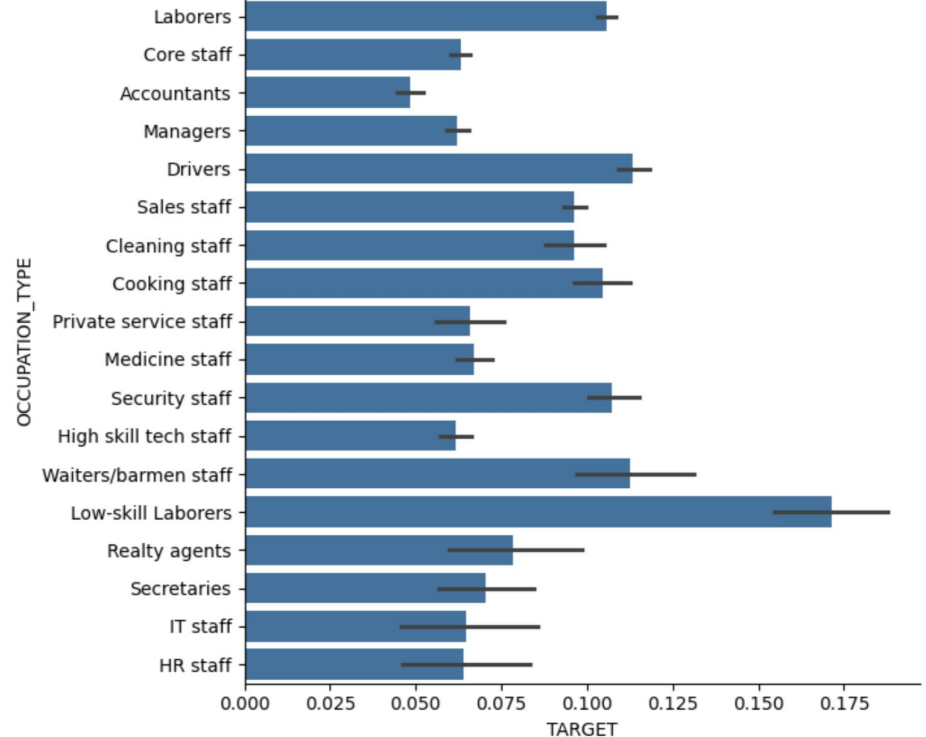


Solution

Providing smaller loans, with longer repayment plans for blue collared workers



<seaborn.axisgrid.FacetGrid at 0x78bad3050320>



Gender

Male

Have the highest default rate



Reasons

Despite income inequality- could indicate a better allocation of funds by women
Difference in the burden of responsibility - could in some cases be higher for men

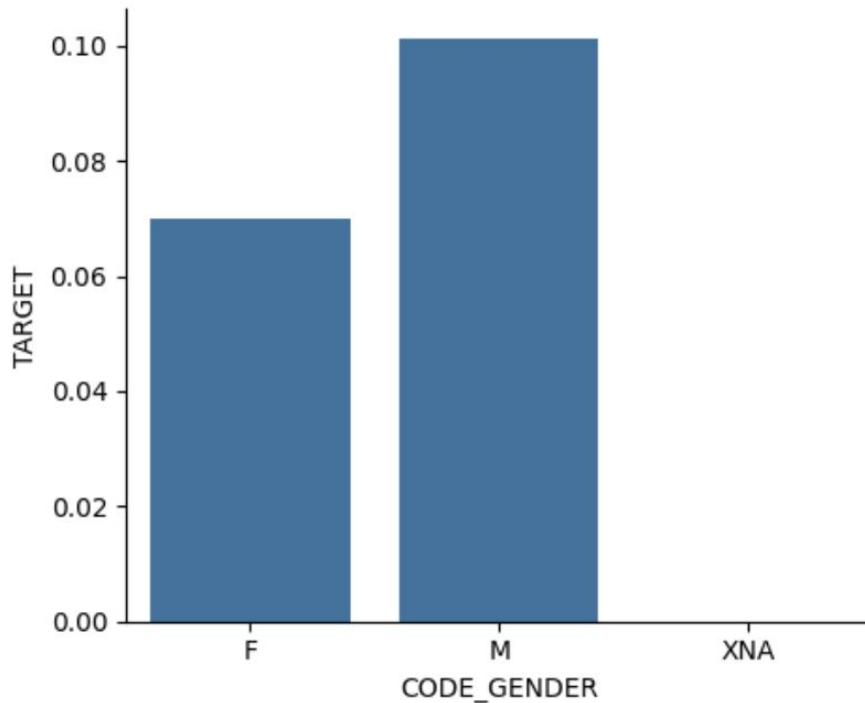


Solution

Providing loans, tailored for women contributing to empowerment (introduction of micro-loans for women in lower income brackets, would lead to escaping poverty cycle and uplift societies



<seaborn.axisgrid.FacetGrid at 0x78bb17ba9550>

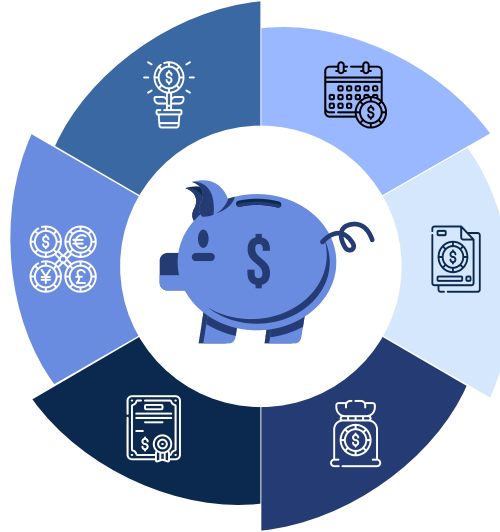


Conclusion- who should banks prioritise

- Higher annuity & income

- Higher education

- Women



Widowed ●

White collared
workers ●

Younger than 30yrs or
retired ●